Experiment on Discharging a Capacitor

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AIM

To study the discharge of a capacitor and determine the time constant τ, and its capacitance C.

LEARNING OBJECTIVES

At the end of this lesson, you should be able to:

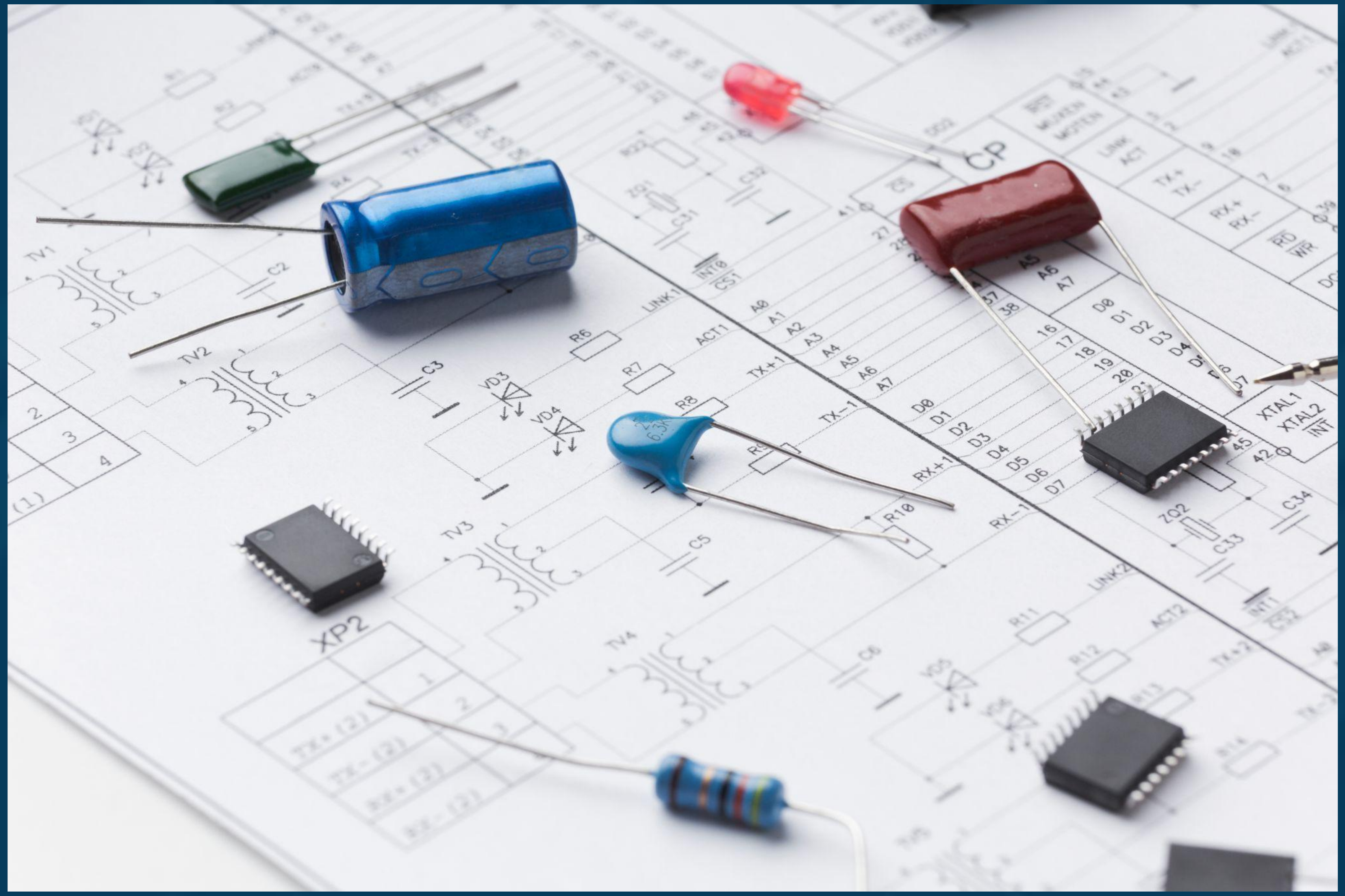
- Study the variation of discharge voltage with time.
- Determine the time constant τ.
- Determine the experimental value of the capacitance of capacitor C.
- Calculate the percentage error for capacitance.

APPARATUS

- Voltmeter.
- Power supply.
- Discharging circuit (Capacitor, Resistor, Switch key).
- Stopwatch.
- Wires.
- Praxilabs virtual laboratory – RC Circuit Discharging

THEORY

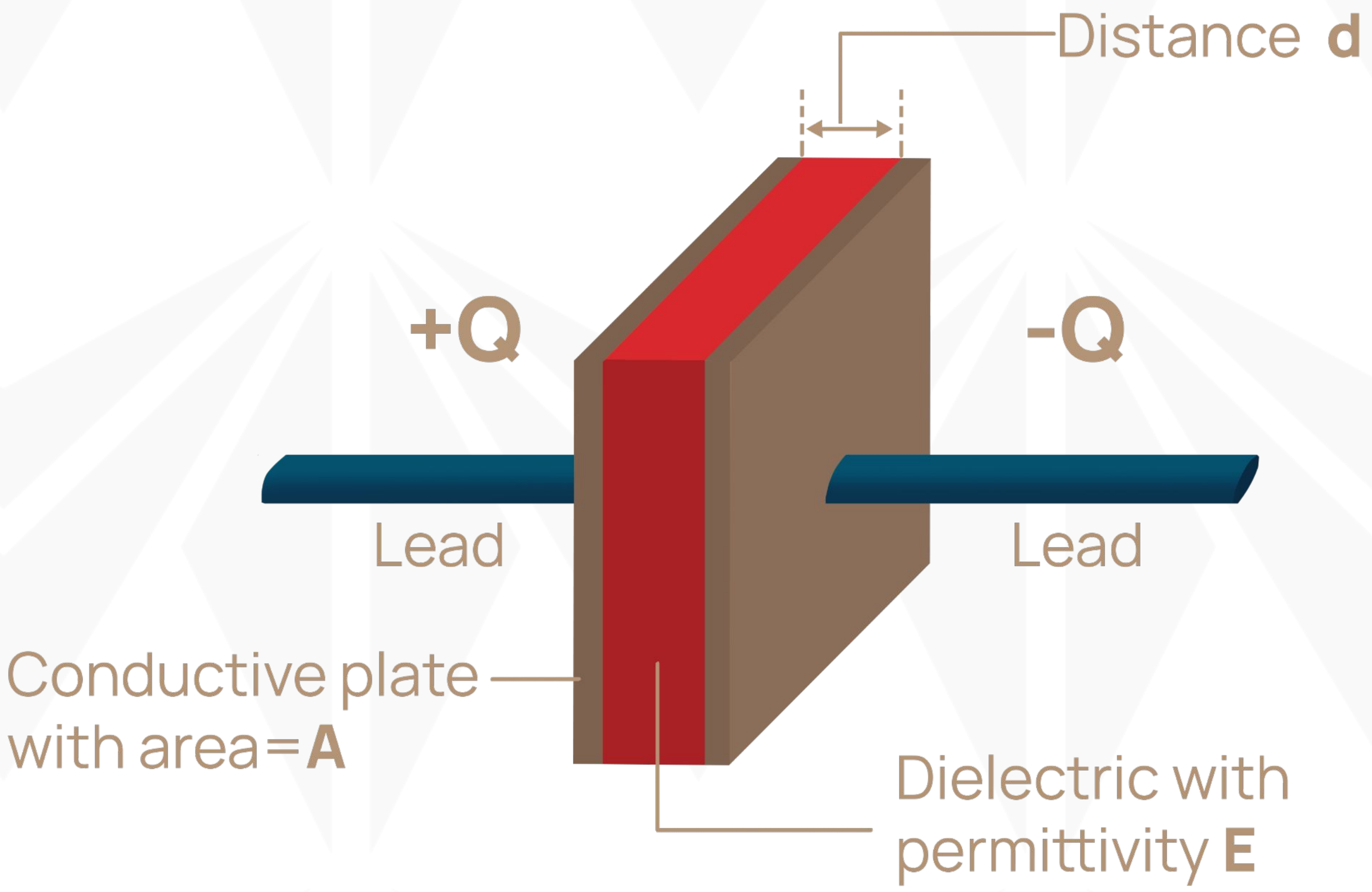
- A capacitor is a device that stores electric energy in an electric field form.
- Capacitors have several uses, such as filters in DC power supplies and as energy storage banks for pulsed lasers.
- Capacitors are electronic components that can pass alternating current (AC) while blocking direct current (DC).



THEORY

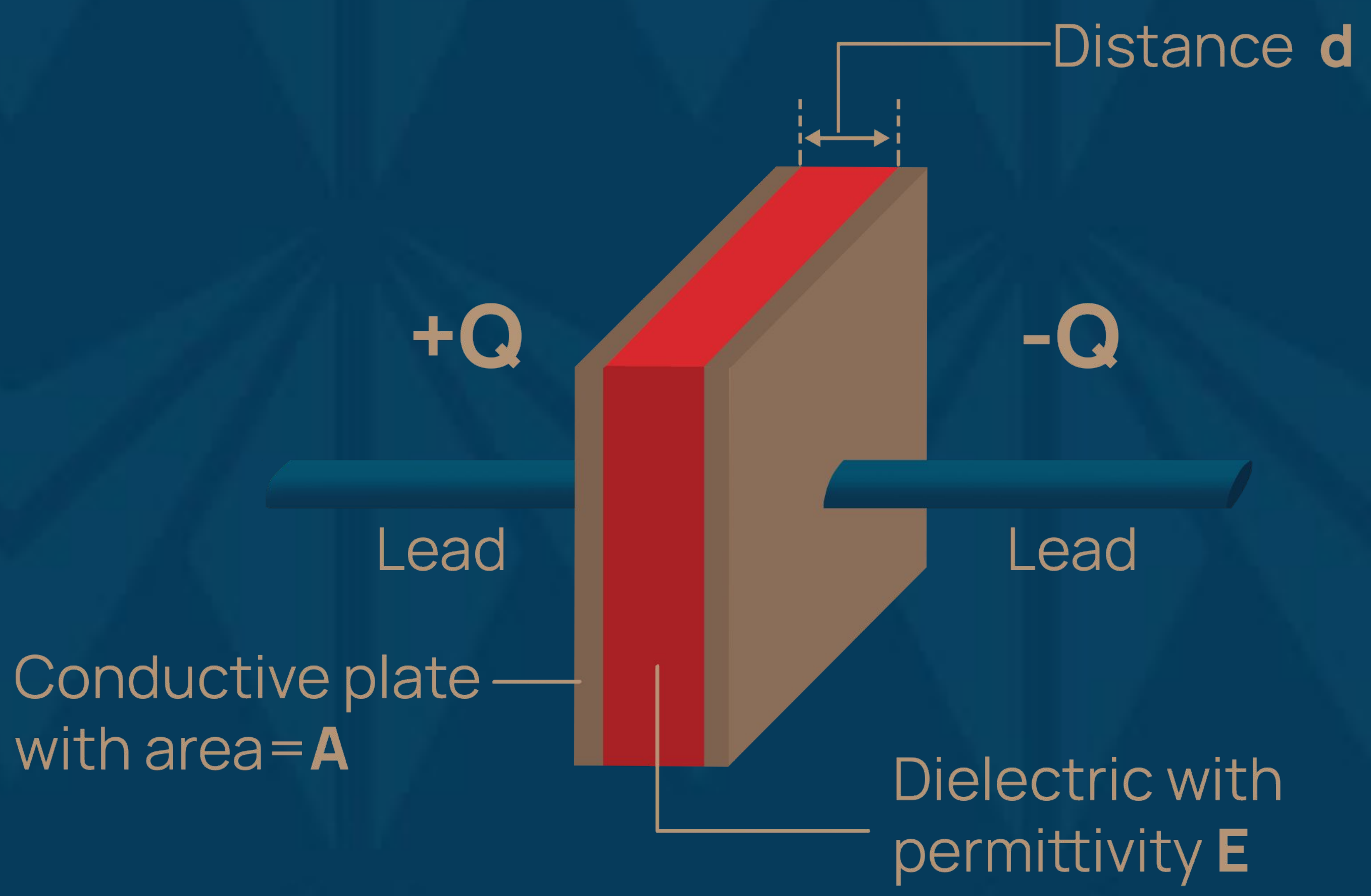
A capacitor consists of two conductors separated by a small distance, d, plate Area, A and dielectric material with permittivity, ε.

$C = \varepsilon A / d$



THEORY

- When the conductors are connected to a charging device (for example, a battery),
- The charge is transferred from one conductor to the other until the difference in potential between the conductors becomes equal to the potential difference between the terminals of the charging device.



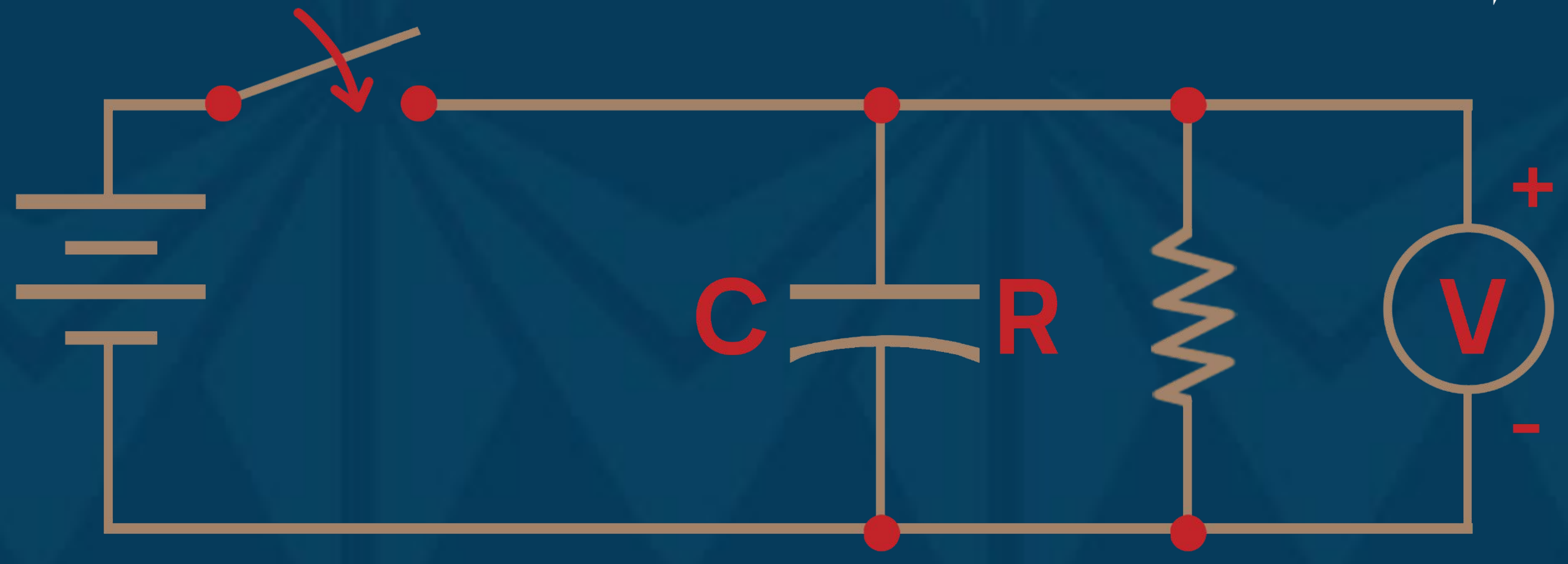
THEORY

- The amount of the charge stored on either conductor is directly proportional to the voltage, and the constant of proportionality is known as the capacitance.

  $Q = CV \quad (1)$
- The charge Q is measured in units of coulomb (C),
- the voltage V in volts (V), and
- The capacitance C in units of farads (F).
- The capacitors are physical devices; capacitance is a property of devices.

THEORY

- In the RC circuit shown:
- Turn on the power supply and turn on the switch key.
- Then turn off the switch key, the voltage on the capacitor decreases exponentially with time (t) according to this equation:
- $V(t) = V_0 \, e^{-t/\tau}$ (2)



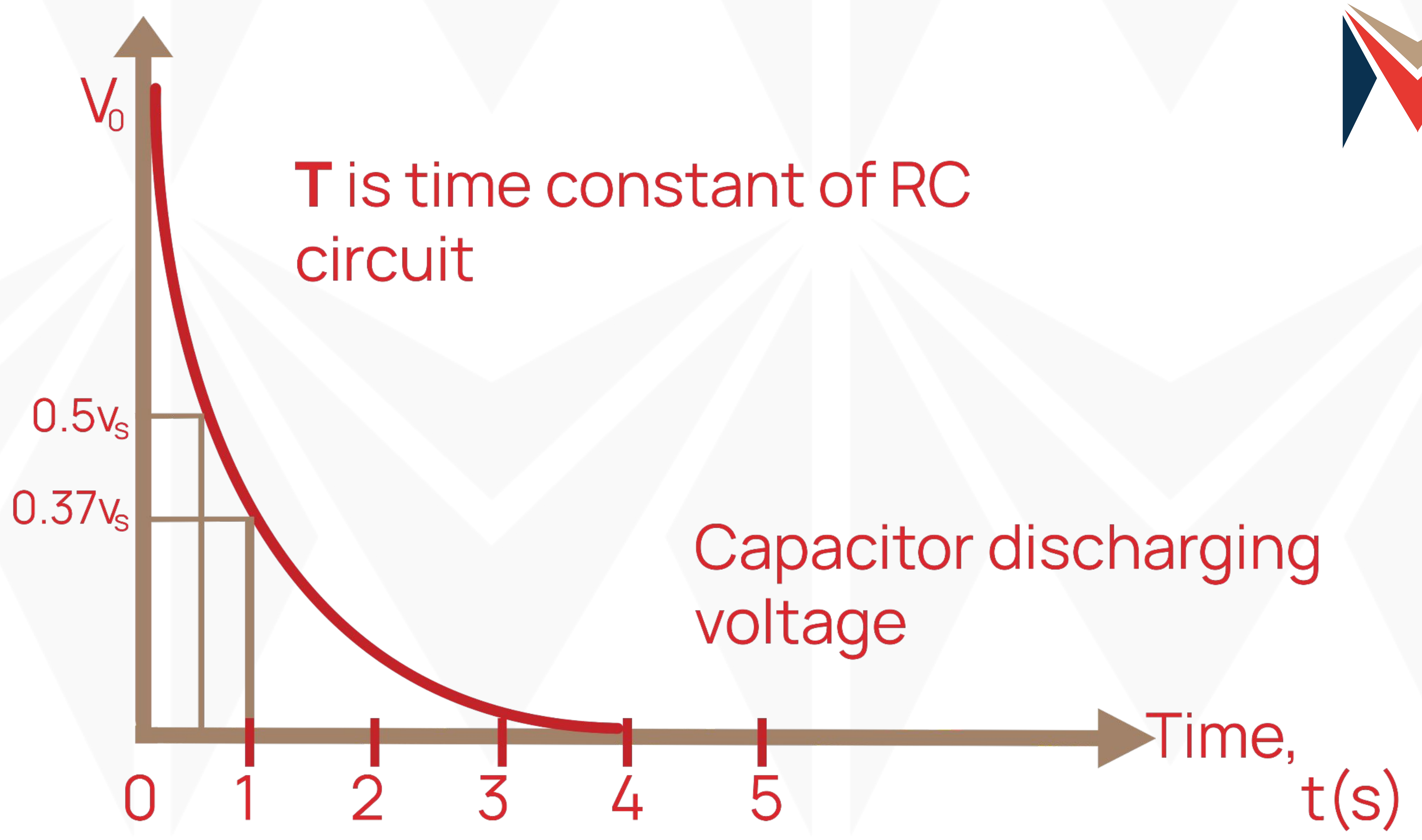
where τ is the time constant, V(t) is voltage on capacitor during discharge at time t.
V_0 is maximum voltage.

THEORY

- The figure shows the exponential curve of a discharging capacitor

- The time constant depends on the resistance (R) and capacitance (C) values in the circuit and is calculated using the following equation:

  $\tau = RC \qquad (3)$



MID-LESSON QUESTIONS

Q1: How do capacitors store electric energy?

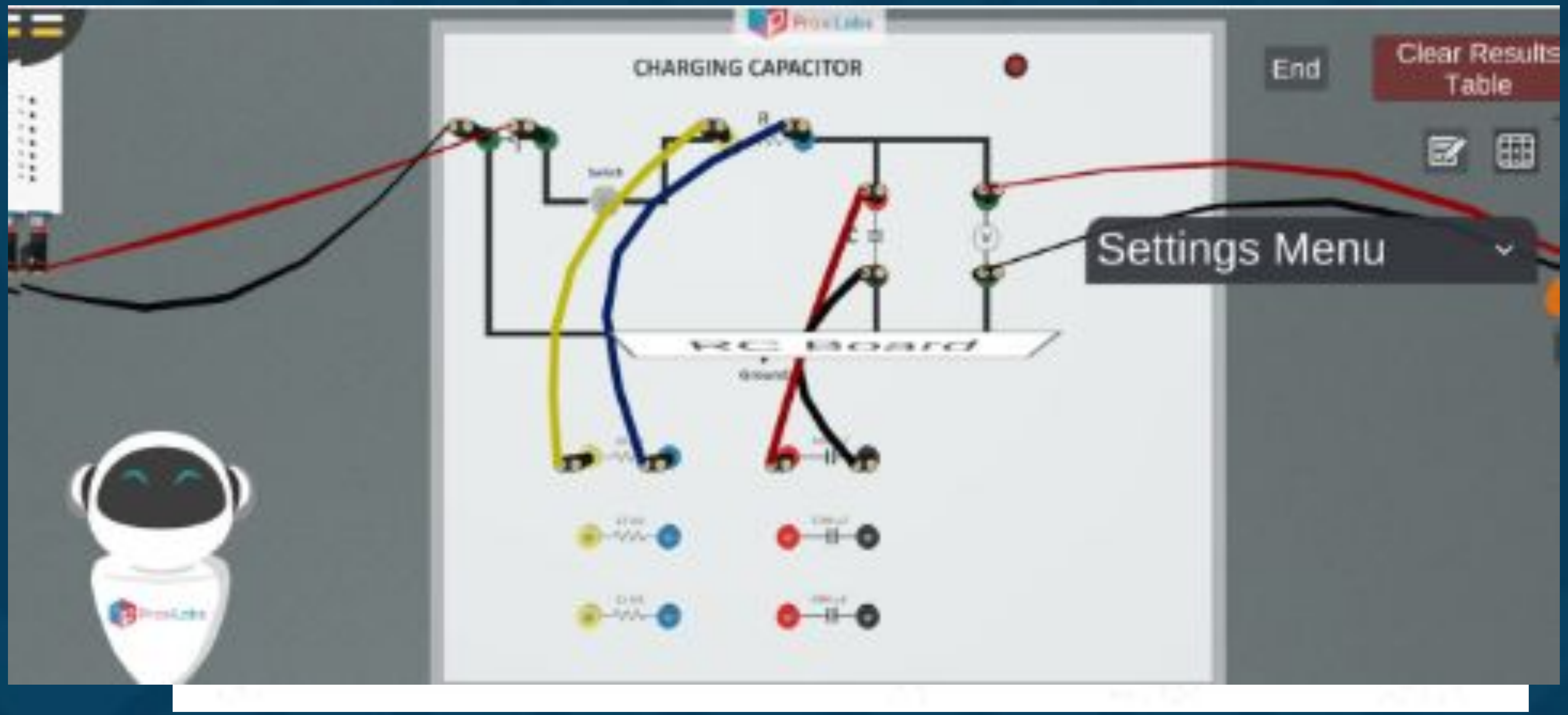
Q2: What are some common applications of capacitors?

MID-LESSON ANSWERS

A1: Capacitors store electric energy in the form of an electric field between two conductors separated by a small distance.

A2: Capacitors are used as filters in DC power supplies and as energy storage banks for devices like pulsed lasers.

PROCEDURE

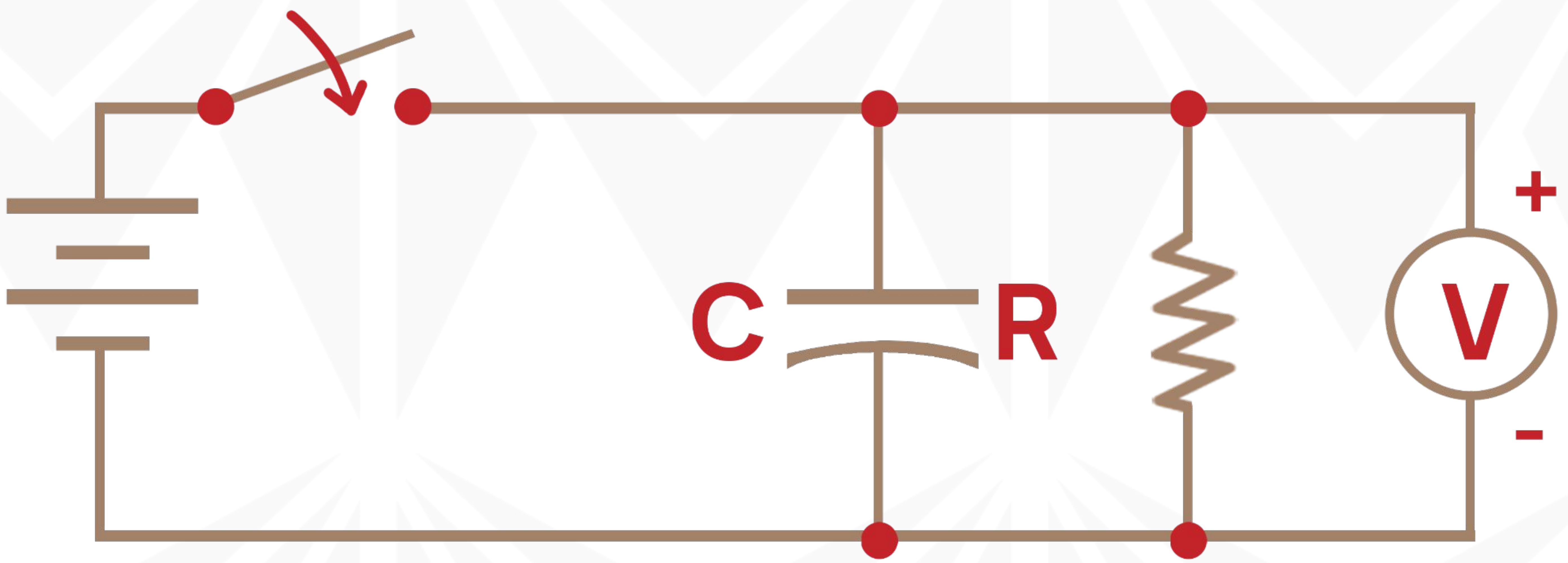


1. Connect the power supply and voltmeter in the discharge circuit then choose a suitable capacitor and resistor to connect them by wires in the discharging circuit as shown.

PROCEDURE

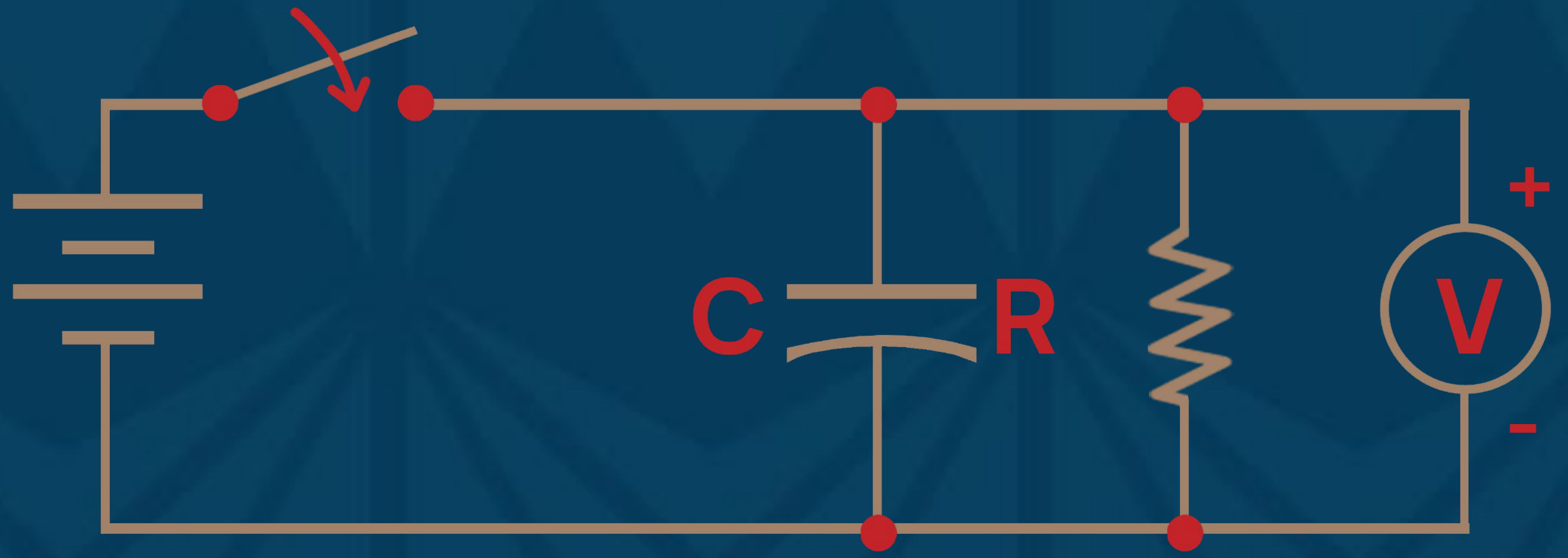
2. Turn ON the power supply and choose a value of output voltage.(For example 15 volts).
3. Turn ON the switch key. Notice: the led has been lighted up.

Note: take 10 measurements at least.



PROCEDURE

4. Wait until the voltmeter reach the maximum value (Approximately the voltage of the power supply).
5. Turn OFF the switch key and with a stopwatch record the voltmeter reading each 10 seconds. You must take at least 7 measurements. Please note that at zero time, the voltmeter reading is maximum.



PROCEDURE

6. You may end the experiment after any trial by clicking the "End" button.
7. Once you click the "End" button, an excel sheet will be downloaded containing the recorded data.

Trial #1:

V_0 (Volt)	11
R (kilo-Ohms)	15
C (microfarads)	2200

t(sec)	V(Volt)
3.16	9.99
10.03	8.12
20.35	5.94
31.33	4.26
40.56	3.22
50.8	2.36
58.12	1.89
70.04	1.32
79.8	0.98
87.59	0.77

Data Analysis

1. Plot the relation between the voltage on the capacitor (on the y-axis) and time, t(s) (on the x-axis).
2. From the curve, get the time constant by putting t=T in equation (2),

$$V(t) = V_0\, e^{-t/\tau} \quad (2)$$

$$V(t) = V_0\, e^{-1} = 0.3679V_0$$

So, the time constant, τ, represents the time it takes for the voltage to reach approximately 0.3679 of its maximum value.

DATA ANALYSIS

3. Draw another relation between the natural log of voltage (*ln V*) (on the y-axis) and the time, t (sec) (on the x-axis).

$V(t) = V_0\, e^{-t/\tau}$

Linearise □ $\ln V(t) = \ln V_0 - t/\tau$

4. Calculate the slope of the line and get the time constant. (We will have two values of the time constant from the two graphs, they have to be equal).

Data Analysis

5. Knowing the value of the resistance, we can get the experimental value of the capacitance using the following equation:

$C = \tau / R$

WALK_THROUGH VIDEO

SUMMARY

The experiment on discharging a capacitor provided valuable insights into the controlled release of electrical energy stored within a capacitor. By setting up circuits, measuring voltages, and rigorously analyzing data, we gained a practical understanding of the discharging process. This experiment not only deepened our comprehension of fundamental physics principles but also underscored the real-world applications of capacitors in electronic circuits.

FURTHER READING

Bueche F. and Hecht E. Schaum's Easy Outlines - College Physics Crash Course. 2000 McGraw-Hill, USA.

Hernández-Hall, C. A., & Wilson, J. D. (2014). Physics Laboratory Experiments. Cengage Learning.

Sang, D., Hewlett, C., Jones, G., Field, S., & Styles, D. (2020). Cambridge International AS & A Level Physics Practical Workbook. Cambridge University Press.

Serway, R.A. and Jewett. J.W. (2014). Physics for Scientists and Engineers with Modern Physics, Ninth Edition. Physical Sciences: Mary Finch.

FURTHER READING

Shukla, R. K., & Srivastava, A. (2006). Practical Physics. New Age International (P) Limited, Publishers.

Young, H. D. (2012). Sears & Zemansky's college physics. 9th ed. Pearson Education, Inc.

http://hyperphysics.phy-astr.gsu.edu

http://www.physicstutorials.org

Thank You